

Biochemistry lab

Experiment (9)



EXPIREMENT 9

CARBOHYDRATE QUALITATIVE ANALYSIS

● Carbohydrates consist primary Of: C, H, O.

● Carbohydrates Formula: $C_n(H_2O)_n$.

Or: $C_nH_{2n}O_n$. $\longrightarrow$ That means that the ratio of C: H: O is **1: 2: 1**

REMEMBER:

Quantitative study: Data are collected by measuring things (دراسة كمية)

Qualitative study: Data are collected by observation (تحليل نوعي لصفات المادة)

Classification of Carbohydrates.

● Carbohydrates will be classified based upon the number of Polyhydroxy aldehyde or ketone (saccharide) units found in their structure.

1-Monosaccharide: one Polyhydroxy aldehyde or ketone (glucose, fructose...)

2-Disaccharide: two saccharides (sucrose, lactose, Maltose...)

3-Oligosaccharide: 3-10 saccharides

4-Polysaccharide: more than 10 saccharides, huge polymers. (Starch, Glycogen...)

"يعني, تم تصنيف الكربوهيدرات بناء على عدد الأليدهايد أو الكيتون المسمى ب saccharide المحتوي على الكثير من مجموعات الهيدروكسيل **OH**, إما يحتوي على saccharide واحد فيعتبر monosaccharide, او اثنتين فيسمى Disaccharide, أو أكثر فيسمى Oligosaccharide".

●Disaccharides and oligosaccharides polysaccharides are bonded together by **Glycosidic Linkage** or **Glycosidic Bond**.

●Glycosidic bond is a covalent bond; it's formed by dehydration (condensation) reaction, by removing one molecule of water, **H** from one and **OH** from another. It is broken by Hydrolysis, giving the carbohydrate H_2O which will give **H** to one and **OH** to another.

REMEMBER: # of Glycosidic linkage = # of H_2O = # of saccharides -1

First classification: Monosaccharides

●monosaccharide consists of only one saccharide molecules, no Glycosidic bond.

●it is further classified based on:

- Number of carbon atoms forming the structure (pentoses 5 or Hexoses 6)

- Functional group present in the structure (ketone or aldehyde)

This leads us to 4 classifications:

1- Aldo pentose (5 carbon atoms with aldehyde functional group)

2- Keto pentose (5 carbon atoms with ketone functional group)

3- Aldo hexose (6 carbon atoms with aldehyde functional group)

4- Keto hexose (6 carbon Atoms with ketone functional group)

Note: The table below must be remembered.

"بيجي عليهم دايما بالامتحان, مثلا مين من التالي يعتبر Aldo hexose؟ أو الرايبوز يعتبر ماذا؟"

Aldo Pentose	Aldo hexose	Keto hexose
Ribose	Glucose	Fructose
Arabinose	Galactose	Tagatose
Xylose	Mannose	
Lyxose	Idose	
	Talose	
	Allose	
	Atrose	

Monosaccharide

basic carbohydrate unit – single polyhydroxy aldehyde or ketone
(الوحدة الاساسية وهي الابسط)

Note: Monosaccharaides can contain either aldehyde group or ketone group.

من المخطط قسمنا السكريات الاحادية على أساس

Aldose	ketose
↓	↓
Like glucose	like fructose
(= bond on C# 1)	(= bond on C# 2)

- 1- عدد الكربونات
- 2- على حسب المجموعة الوظيفية
, فمثلاً الفركتوز الرابطة الثنائية
بتكون على الكربونة الثانية فهو
كيتون , أما الجلوكوز الرابطة
الثنائية على الكربونة الاولى لهذا
فهو الديهايد

Any saccharide has two forms

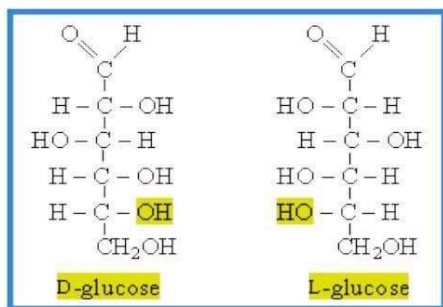
→ openchain (fischer projection)

→ cyclic chain(haworth projecton)

1) Open chain (Fischer projection)

- Exists only at low concentration

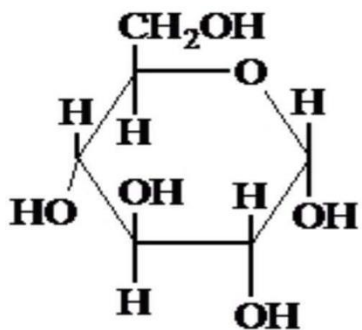
Isomers of glucose



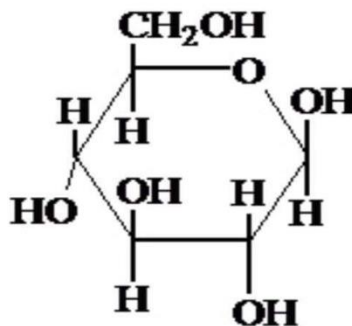
- If the highest numbered chiral carbon has OH on right → D-glucose.
- If the highest numbered chiral carbon has OH on left → L-glucose.

2) Cyclic chain (Haworth projection)

- Only the cyclic form is found in the solid state.



alpha-glucose



beta-glucose

- Cyclic chain form when C #1 binds with C #5
- We can determine if it is α -glucose or β -glucose from the anomeric Carbon
- If OH below the ring → α -glucose
- If OH above the ring → β -glucose
- Anomeric carbon → the carbon that is bounded to O (C #1 in glucose. It can be C #2 in other sugars.)

Disaccharides

They are carbohydrates which upon hydrolysis (reaction with water) gives two monosaccharide structures.

Binding of two monosaccharide requires removal of 1 H₂O

EXAMPLPLES:

Sucrose: glucose + fructose

Lactose: glucose + galactose

maltose: 2 glucose.

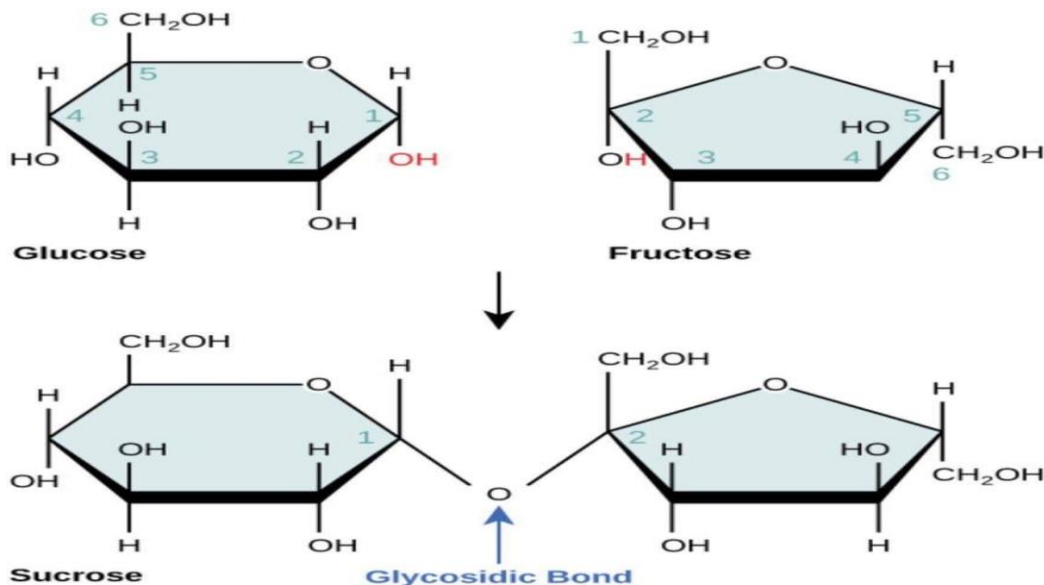
- More explanation:

1) Sucrose: linkage of β -fructose and α -glucose (D-glucose and D-fructose).

The two sugar units are covalently joined (Glycosidic bond),

through an oxygen at carbon 1-2 on α glucose and β fructose , respectively.

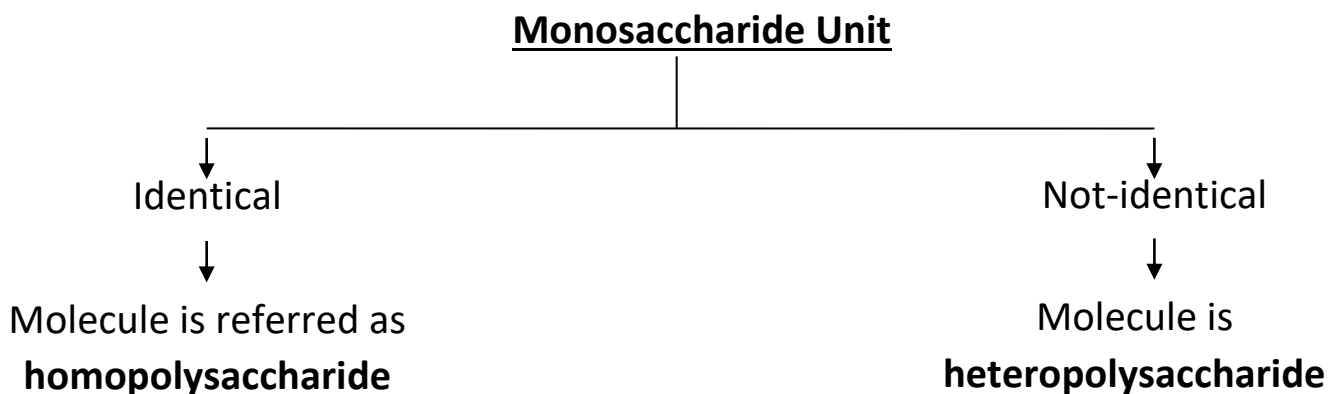
(يتكون السكروز عن طريق الاكسجين الذي يربط الكربونة رقم 1 من ألفا جلوكوز والكربونة رقم 2 من بيتا فركتوز فتتكون الرابطة) ($\alpha 1 \rightarrow \beta 2$)



- 2) Maltose: contains two D-glucose units joined by an α (1 $\rightarrow$ 4) Glycosidic bond
- 3) Lactose: consists of D-galactose and D-glucose joined by β (1 $\rightarrow$ 4) linkage
- The **Glycosidic linkage** are destroyed when disaccharides undergo hydrolysis back to the individual monosaccharide

Polysaccharides:-

Consist of large number of Monosaccharides, joined together by Glycosidic linkages.



Example:-

1. Starch

→ Amylose (linear)

→ Amylopectin (branched)
2. Glycogen
3. Cellulose

more explanation//

Starch

- Is a Homopolysaccharide synthesized by plants for storage of α -D-glucose units.
- Serves as a source of carbohydrates in animal diets.

Mixture of two different polymeric structural units.

Amylose (linear)

Glucose is joined by α (1 $\rightarrow$ 4) Glycosidic bonds.

(Longer)

Amylopectin (Branched)

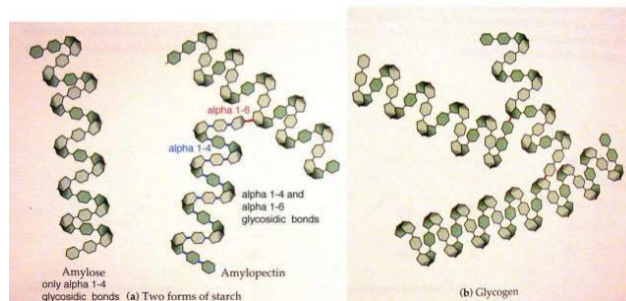
Glucose is joined in linear chains by α (1 $\rightarrow$ 4) bonds, and by α (1 $\rightarrow$ 6) at branch points.
(التفرع كل 30-24 جلوكوز)

Glycogen

- storage of polysaccharide in animals

- Similar to amylopectin, but larger.

Starch Vs Glycogen...



Carbohydrates chemical reactions:-

* **Reducing sugar:** is any sugar that is capable to act as a reducing agent because it has a free aldehyde group or a free ketone group (the ring structure of sugars are able to open at anomeric carbons to form the open-chain aldehyde or α -hydroxyketone atom), which undergoes oxidation, and reduces another thing, such as benedicts' reagent

"أي أن ال Reducing sugars هي السكريات القادرة على العمل كعامل مختزل بسبب وجود الألديهيد و الكيتون، حيث أن الحلقة عند وضعها في المحلول تفتح الحلقة عند ال anomeric carbons و تكوّن سلسلة مفتوحة، فيظهر ال الديهايد و الكيتون و يحدث لهم أكسدة، بالمقابل تختزل شيئاً آخر."

Important:

1. When anomeric ring carbon is involved in a glycosidic linkage → the ring is locked and can't open for oxidization to occur.
2. All monosaccharides are reducing sugars → because no glycosidic linkages exist in their structures; that means that all the anomeric carbons aren't involved in Glycosidic linkage.
3. Disaccharides → Only Maltose and Lactose are reducing sugars → because only one of the anomeric carbons (C_1) is involved in a glycosidic linkage.

Succrose → non-reducing sugar → since both C_1 and C_2 anomeric carbons participate in the glycosidic linkage.

4. Polysaccharides → non-reducing sugars → because of their size and complex structure.

"المالتوز يربط الكربونة 1 في الجلوكوز مع الكربونة 4 في الجلوكوز، الأولى anomeric والثانية لا. بالمقابل، السكروز يربط الكربونة 1 في الجلوكوز مع الكربونة 2 في الفركتوز، وكلاهما anomeric carbons."

In this Experiment :

-we want to examine carbohydrates in different ways (qualitative test)
(differentiate between sugars) using 5 tests.

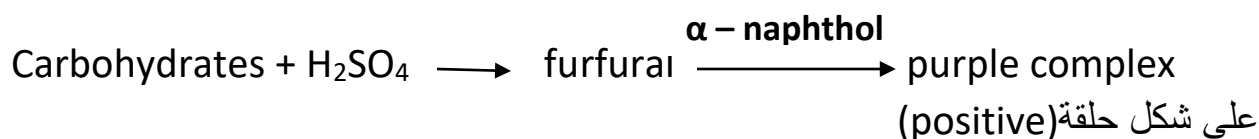
1) molish's test :

Purpose: it is considered a general test for all carbohydrates (to determine if the compound is carbohydrates or not)

In other meaning: to test for the presence of carbohydrate.

Reagent: α - naphthol in 95% ethanol added to its concentrated H_2SO_4

Result: purple ring



Notes:

- 1) the addition of H_2SO_4 must done while the tube is bent with an angle 45° not 90°
- 2) work under fume hood
- 3) monosaccharides react more quickly than di and polysaccharide
- 4) H_2SO_4 is for dehydration and making furfural, α - naphthol is for condensation and forming of purple complex

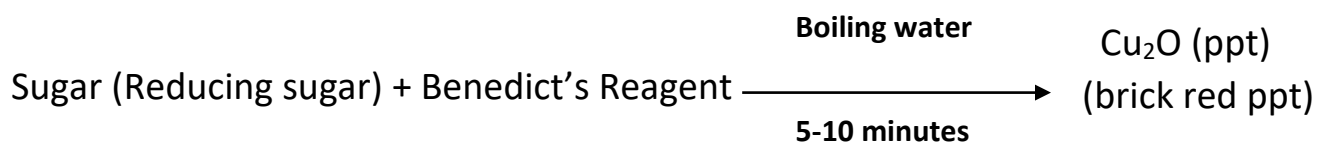
"يعني, آلية التفاعل تكون بالبداية تفاعل تجفيف عن طريق الحمض, ثم تفاعل
condensation عن طريق α - naphthol لتكوين لون بنفسجي."

2) Benedict's test

Purpose: to determine if the carbohydrate contains reducing agent.

Benedict's Reagent: - (Cu^{+2} / NaOH base).

Results: - Red precipitate.



Notes:

1) Sugar $\longrightarrow$ will be oxidized while it will reduce Benedict's reagent (back to page 8 to remember the reducing sugars.)

2) The reduced form of Cu^{+2} is Cu_2O . Cu^{+2} becomes Cu^{+1}

3) All monosaccharides are reducing agents.

-Disaccharides are either reducing or not reducing (Sucrose is not reducing while Lactose + Maltose are reducing)

-Polysaccharides are non-reducing in general.

4) Disadvantage: - also other compounds like Aldehydes, Formic acid and phenols will reduce Benedict's reagent and give positive results.

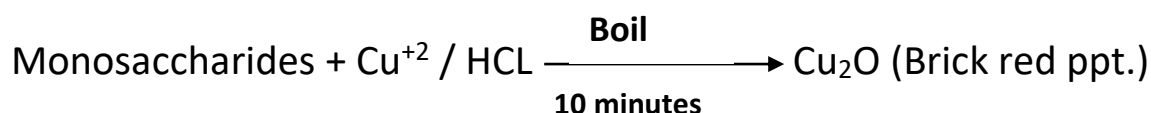
"وهذا الشيء سلبي، لاننا لن نستطيع تمييز الكربوهيدرات من بين هؤلاء المركبات عن طريق هذه التجربة."

3) Barfoed's Test

Purpose:- To determine which is faster reducing agent and which is slower (reducing Monosaccharides).

Barfoeds solution: - (CuSO_4 in acidic medium.)

Results: - Red / brown of (Cu_2O) ppt.



Notes :-

1 -prolonged boiling point (10 min) will hydrolyze disaccharide to give positive results whereas the appearance of ppt in Monosaccharides is within (2 -3 minutes)

And this is a positive thing; it gives us the ability to differentiate between Monosaccharides and Disaccharides.

"يعني, السكريات الأحادية تحتاج لدقيقتين والثنائية تحتاج لعشرة, وهكذا نميز بين الأحادية والثنائية, عن طريق السرعة."

2 -The precipitate is less dense than that in Benedict's so leave it to allow the ppt settle.

3 -Barfoed's test looks very similar to Benedict's test, but the difference is that barfoed's test in acidic medium whereas benedict's test in basic medium.

4) Bial's Test

purpose: -To differentiate pentoses and hexoses (esp:- Aldopentoses) structure .

Dehydration reagent

condensation reagent

Bials reagent: - Concentrated HCl +Fe⁺³ iron (III) + orcinol.

Result: - Blue green color (with Aldopentoses).

Aldopentoses (ribose) + Bial's reagent $\xrightarrow[5\text{ minutes}]{\text{Boil-heat}}$ Blue green color.

Notes :-

1) concentrated acid is used to dehydrate the sugars which produce colorless dehydration products .

2) this dehydration products then react with condensation (orcinol) reagent to form color $\longrightarrow$ positive results.

- During the reaction, the pentose reacts with HCl and gives Furfural (cyclic aldehyde) which then condenses with orcinol and gives a color.

3) This test is specific for aldopentosis but it's not absolute specific. prolonged heating of aldohexoses will give positive result though.

4) Orcinol (3,5-dihydroxytoluene)

حلقة بنزين عليها 2 OH و CH₃

5) All other colored products are negative indications for the presence of pentose.

"يعني, اللون اللي بيهمنا هو ال blue green فقط, أي لون آخر يعتبر negative indication."

6) bial's test is like molish's test; dehydration then condensation, but the condensation reagent differs.

5) Seliwanoff's Test

Purpose: - To differentiate between ketohexoses and aldohexoses (Esp. ketohexoses).

Seliwanoffs reagent: - HCl (dehydration agent) + resorcinol (condensation agent) (Boil).

Results: - **Red – Bright** complex (with ketohexoses).

Ketohexoses (Fructose) + HCl+ resorcinol $\xrightarrow[3-5 \text{ minutes}]{\text{Boil}}$ **Bright red complex**

Notes: -

1) If prolonged period of boiling we will have +ve results (**pink** color) in Aldohexoses .

2) Resorcinol is meta-dihydroxybenzene

حلقة بنزين عليها 2OH

3) Ketohexoses are dehydrated by dehydrating agent to 5 –hydroxymethyl furfural which undergoes condensation with resorcinol to form a red color.

4) all selwinoff and bial and molish tests have the same mechanism (dehydration then condensation) but they differ in dehydration and condensation agents.

Selwinoff's condensation agent $\longrightarrow$ resorcinol

Bial's condensation agent $\longrightarrow$ orcinol

Molish's condensation agent $\longrightarrow$ α - naphthol.

Methods

Molish's test:

- 1-add about 1 ml carbohydrates
- 2-add 2 drops of the molish's reagent and mix gently
- 3-add 0.5 ml of H_2SO_4 down the side of the tubes, purple ring will form.

Benedict's Test:

- 1-Add about 0.5 ml sugar.
- 2-Add 2.5 ml Benedict's reagent.
- 3-Place the tube in a boiling water bath for 5-10 minutes.
- 4-Observe the color of the solution and precipitate.

Seliwanoff's test:

- 1-Add about 0.5 ml sugar.
- 2-Add 2ml seliwanoff's reagent
- 3-Place the tube in boiling water bath for 3-5 minutes.

Barfoed's test:

- 1-Add about 0.5 ml sugar.
- 2-Add 2.5 ml Barfoed's reagent.
- 3-Place the tube in boiling water bath for 10 minutes.

Bail's test:

- 1-Add about 0.5 ml sugar.
- 2-Add 2.5 ml Bial's reagent.
- 3-place the tube in boiling water bath for 5 minutes.

Important note:

succrose may dissociate and give glucose (aldohexose) and fructose (ketohexose)

So, succrose may give +ve result in seliwanoff's and burfoed's tests

Test	Reagent	Used for	Result	
Molish	α -naphthol H_2SO_4	All carbohydrates	Purple ring	
Bendict	Cu^{+2} in basic medium(blue)	Reducing sugars	Red,brown of Cu_2O ppt	
Barfoed	Cu^{+2} in acidic medium	Monosharide (reducing)	Red,brown of Cu_2O ppt	
Seliwanoff	HCl / resoricinol	Ketohexose (fructose)	Bright red	
Bail	HCl/ Fe^{+3} / oricinol	Aldopentose (arabinose)	Blue-green	

Questions

1) The general formula of carbohydrates :

- a) $(CH_2O)_n$
- b) $(C_4H_2O)_n$
- c) $(C_6H_{12}O_5)_n$
- d) $(C_2H_2O)_n COOH$

2) A positive Benedict's test is not given by

- a) Sucrose
- b) Maltose
- c) Lactose
- d) Glucose

3) In amylopectin the intervals of glucose units of each branch is

- a) 10-20
- b) 24-30
- c) 30-40
- d) 50-60

4) Starch is a:

- a) Polysaccharide
- b) None of these
- c) Disaccharide
- d) Monosaccharide

5) In glucose the orientation of the $-H$ and $-OH$ groups around the carbon atom 5 adjacent to the terminal primary alcohol carbon determines

- a) α and β anomers
- b) D or L series
- c) Dextro or levorotatory
- d) Epimers

6) Two monosaccharide are joined by :

- a) peptide bond
- b) phosphodiesterbond
- c) glycosidic bond
- d) Hydrogen bond

7) Which of the carbohydrate molecules are non-reducing agents

- a) Maltose
- b) Sucrose
- c) Lactose
- d) Glucose

8) The general test for all carbohydrates :

- a) Molisch's test
- b) Benedict's test
- c) Barfoed's test
- d) Bial's

9) Which of these are correct combinations of monosaccharides to form disaccharides?

- i) Glucose + Glucose = Maltose
- (ii) Glucose + fructose = Lactose
- (iii) Glucose + fructose = Sucrose
- (iv) Glucose + galactose=Lactose

- a) (i),(ii),(iii) only
- b) (ii) only
- c) (i),(iii),(iv)only
- d) All of the above

10) What test is used for reducing sugars:

- a) Biuret test
- b) Ninhydrine test
- c) Glucose Oxidase system

d) Benedict's test

11) What are the two structures making up starch:

- a) Amylose and cellulose
- b) Amylose and amylopectin
- c) Amylopectin and cellobiose
- d) Cellobiose and cellulose

12) Is Fructose a non-reducing sugar and why :

- a) No, because it becomes reduced to a brick red colour with benedict's solution
- b) Yes, because it becomes reduced to a brick red colour with benedict's solution
- c) Yes, because it isn't reduced in benedict's solution, colour remains blue.
- d) No, because it isn't reduced in benedict's solution, colour remains blue.

13) What is the name of the bond formed between glucose and galactose monosaccharides to produce the disaccharide lactose?

- a) Alpha 1,2 glycosidic bond
- b) Beta 1,4 glycosidic bond
- c) Alpha 1,4 glycosidic bond
- d) Beta 1,2 glycosidic bond

14) Choose a sugar out of the following that is non reactive to Seliwanoff reagent :

- a) Sucrose
- b) Fructose
- c) Inulin
- d) Ribose

15) Which test can be undertaken to differentiate between Glucose and Fructose?

- a) Benedict
- b) Barfoed's
- c) Seliwanoff
- d) Molisch

16) Molish reagent :

- a) Alpha-naphthol, H₂SO₄
- b) HCl, Fe³⁺/orcinol
- c) Cu²⁺ in basic medium
- d) Non of the above

17) ketohexose + HCl /resorcinol (boil) :

- a) blue green color
- b) bright red complex
- c) pink color
- d) purple color

1	2	3	4	5	6
A	A	B	A	B	C
7	8	9	10	11	12
B	A	C	D	B	B
13	14	15	16	17	
B	D	C	A	B	



DONE BY: Hussein Alsaleh, Hisham Darwish, Dina Nasser Eddine, Lynne Abdullah, Abed Al Hadi Raed, Omar Al-Ali, Danah Alghanim and Ayham Haddad.

